Malik's Mural: Multiplying Big Numbers

Lesson 1 · Wednesday, October 21, 2026 · Your task list says which page to do and when.

Part 1: Word Power

Write the best word from the bank in each blank. One word is extra.

Word bank: factor product partial product estimate regroup quotient

1. In 7 × 9 = 63, the number 7 is a _____.

2. The number 63 is the _____ of 7 and 9.

3. Before I multiply, I round the factors to make an _____.

4. When 8 × 4 = 32 in the ones column, I write the 2 and _____ the 3.

5. In 18 × 13, the number 54 is one _____ _____.

Part 2: Estimate first

Round each factor to the nearest ten. Then multiply.

6. 39 × 21 is about _____ × _____ = _____

7. 57 × 32 is about _____ × _____ = _____

Part 3: Area model

Across the top: 39 = 30 + 9. Down the side: 21 = 20 + 1. Fill all four boxes, then add them.

8. 39 × 21

	30	9
20		
1		

Total: _____ + _____ + _____ + _____ = _____

Part 4: Find the mistake

Find the mistake. Explain it. Then write the right answer.

9. Malik wrote: 27 × 14 = 108 + 27 = 135. What went wrong? What is the right answer?

Part 5: Stack it

Use the standard algorithm. Ones row, then write the 0 for the tens row, then add. Check with your estimate. Use grid paper if you need more room.

10. 57 × 32

11. 146 × 23

Part 6: Mural problem

Estimate first. Then solve on paper.

12. Malik's mural has 15 rows of 48 tiles. How many tiles in all? Estimate: _____

13. Is your answer to #12 reasonable? Use your estimate to explain.

KCE PATHWAYS · GRADES 4–5 · UNIT 2: THE NEXT LEVEL · WEEK 6 · MATH

Malik's Mural Project

Name: _____ Date: _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Before the First Tile

Malik's art class is about to start a giant mural project. Before you meet him, warm up on the exact math skills he'll need today.

1 Match each word to its meaning. Draw a line, or write the matching letter next to each word.

___ **standard algorithm**	**A.** A smart, quick guess made by rounding first.
___ **partial product**	**B.** The stacked, step-by-step way of multiplying larger numbers on paper.
___ **regroup**	**C.** A number being multiplied.
___ **product**	**D.** Carrying extra value into the next place value column.
___ **factor**	**E.** One piece of a multiplication problem, found before adding all the pieces together.
___ **estimate**	**F.** The answer to a multiplication problem.

2 Preview: multiply 14 × 3 using partial products. Break 14 into 10 + 4, multiply each part by 3, then add. Show every step.

3 Have you ever helped with a big group project, like a mural, poster, or art display? Write 1–2 sentences about it.

KCE PATHWAYS · GRADES 4–5 · UNIT 2: THE NEXT LEVEL · WEEK 6 · MATH

Malik's Mural Project

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Counting Every Tile

Great work on today's lesson! Now show off what you learned about the standard algorithm, partial products, and regrouping.

1. The gym mural was such a hit that Malik's class is designing a SECOND mural for the cafeteria wall: 32 tiles across and 26 tiles down. Use the standard algorithm to find how many tiles it needs. Show every step, including both partial products.

2. Each jar of paint covers 148 tiles. Malik has 6 jars. How many tiles can all 6 jars cover? Use the standard algorithm and show where you regroup. Is that enough paint for the cafeteria mural?

KCE PATHWAYS · GRADES 4–5 · UNIT 2: THE NEXT LEVEL · WEEK 6 · MATH

Malik's Mural Project

Name: _____ Date: _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON · CONTINUED

Counting Every Tile

3 Gold border tiles come in packs of 28, and the class needs 19 packs. First, ESTIMATE the total number of gold tiles by rounding each factor to the nearest ten. Then use the standard algorithm to find the exact product. Was your estimate higher or lower than the exact answer? Why?

4 Malik's classmate multiplied 24 × 18 and got 216. Malik checked his work: the first partial product (24 × 8 = 192) was right, but the second one was written as 24 instead of 240. Explain in 1–2 sentences what mistake his classmate made, and find the correct product.

★ **Challenge:** Design your own tile mural! Choose a 2-digit number of tiles across and a 2-digit number of tiles down. Use the standard algorithm to find the total number of tiles, then estimate to check that your answer makes sense.

Looking Ahead: Next week we divide BIG numbers using 2-digit divisors — Deja needs help splitting her class's food drive cans evenly among 24 pantries! Bring this sheet back to your next lesson.

KCE PATHWAYS · GRADES 4–5 · UNIT 1: FIRST STEPS · WEEK 3 · READING

Making the Case

Name: _____ Date: _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Before the Slammed Door

Before you meet Marcus, warm up your reading brain — and your handwriting — on last week's skill AND today's new ones.

REV Quick Review from Week 2: In your own words, what does PERSPECTIVE mean? Write your answer in a full sentence.

1 Match each word to its meaning. Write the matching letter next to each word.

___ **cause**	**A.** Writing meant to convince someone
___ **effect**	**B.** The reason something happens
___ **persuasive**	**C.** What happens as a result

2 Handwriting Focus: Copy this sentence in your best cursive or print: “Calm facts often persuade better than loud feelings.”

3 Think of a time you tried to convince a parent or adult of something. Write 2-3 sentences about what you said and what happened.

KCE PATHWAYS · GRADES 4–5 · UNIT 1: FIRST STEPS · WEEK 3 · READING

Making the Case

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

The Deal

Marcus and his dad reached a compromise. Now show what you learned about cause/effect, persuasion, and theme — with your very best handwriting.

1. In 3-4 sentences, explain the CAUSE and EFFECT chain that led from the canceled trip to the final compromise. Use complete sentences and your neatest handwriting.

2. Find ONE physical detail from the story that shows Marcus is nervous, WITHOUT the text stating the feeling directly. Copy it exactly, word for word, in your best handwriting.

3. In your own words, write what this story's THEME is. Write in full sentences.

★ **Challenge:** Write a short PERSUASIVE paragraph (4-5 sentences) making a real case to a parent or guardian about something you'd like — using at least one piece of real "evidence" to support your case. Use your best cursive or print — this is a handwriting sample, so take your time!

Handwriting Check: Before turning this in, look back over everything you wrote today. Are your letters sitting on the lines? Are your capital letters and periods where they should be?

Looking Ahead: Next week, a new story and a new challenge — bring this sheet back to your next lesson!

Name: _____ Date: _____ / _____ / _____

Graph Paper Practice · Page 1

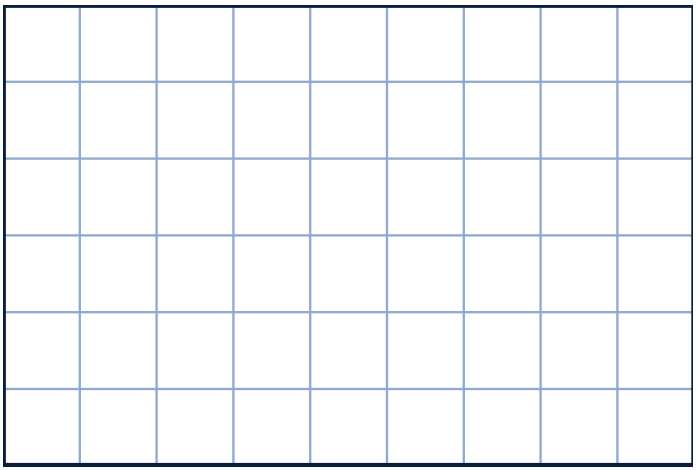
Friday. Work each problem on the grid: **one digit in each square**. Estimate first. Then multiply with partial products, add them, and compare with your estimate.

1 57 × 34

My estimate: ☐

Round, then multiply: 60 × 30

Close to my estimate? **Yes · No**

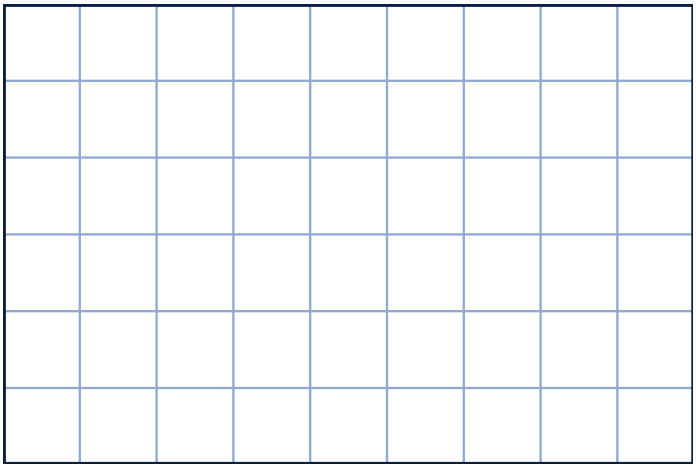


2 83 × 46

My estimate: ☐

Round, then multiply: 80 × 50

Close to my estimate? **Yes · No**

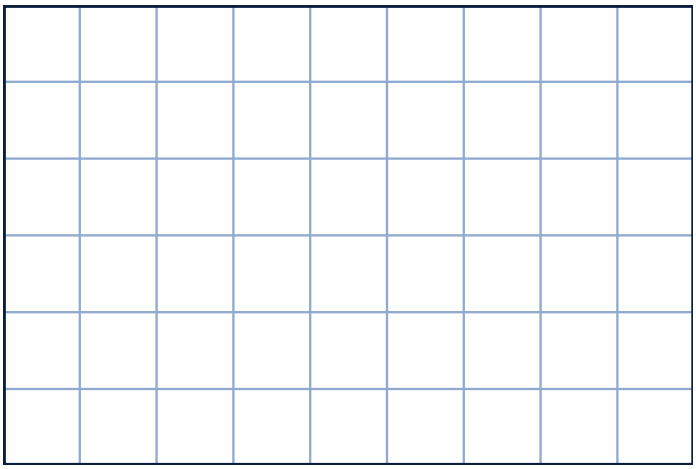


3 124 × 23

My estimate: ☐

Round, then multiply: 120 × 20

Close to my estimate? **Yes · No**

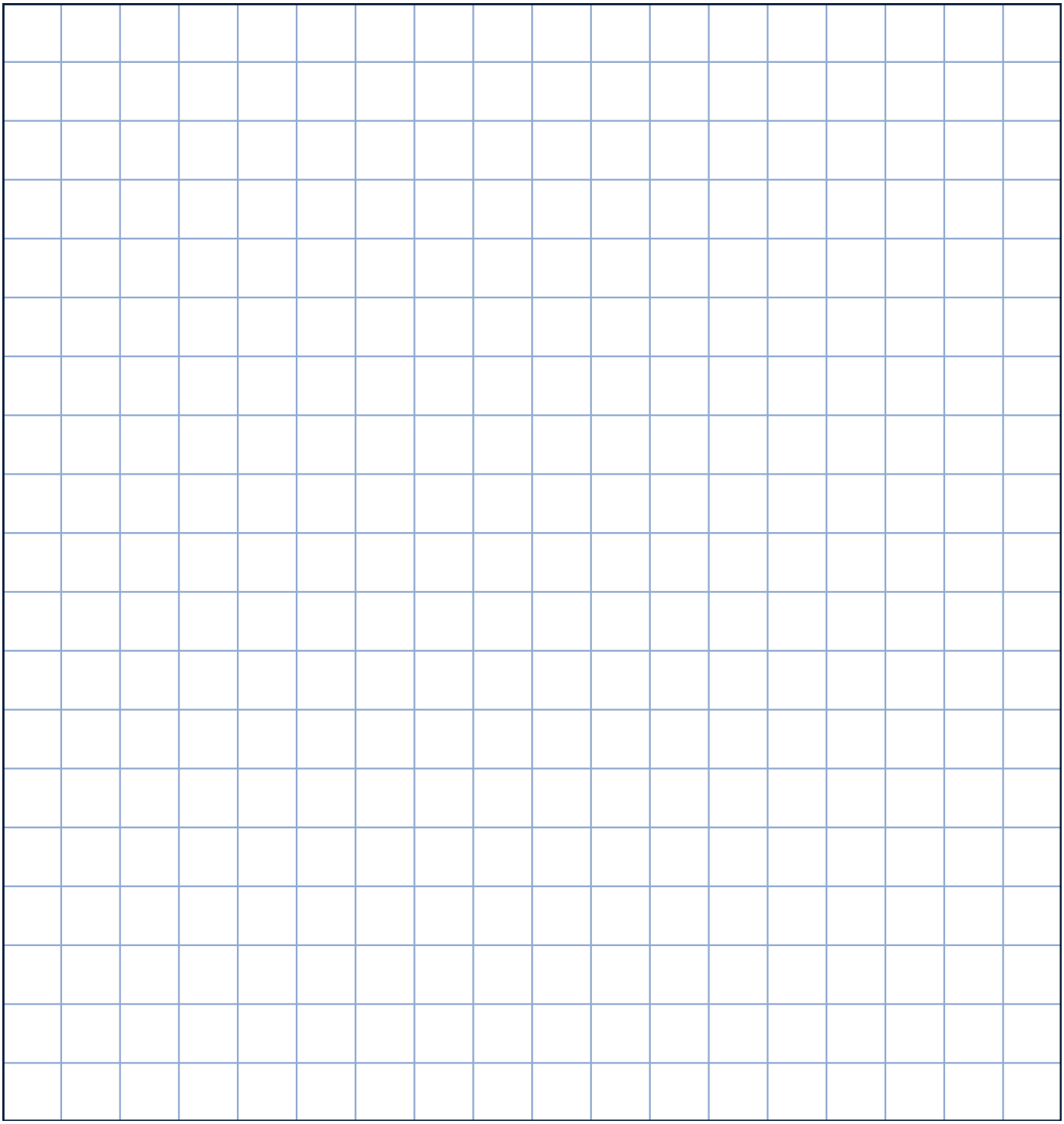


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Name: _____ Date: _____ / _____ / _____

Graph Paper Practice · Page 2

Saturday. Redo page 2 of your Lesson 1 Packet here. Copy each problem, then work it with **one digit in each square** so every column lines up.



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